

Haoyu Liu

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EDUCATION

University of Southern California | Los Angeles, CA Aug 2023 - May 2026

- Master of Science in Computer Science | GPA: 3.71/4.00

University of Illinois, Urbana-Champaign | Champaign, IL Aug 2019 – May 2023

- Bachelor of Science in Computer Science & Linguistics | GPA: 3.91/4.00

Related Courses: *Data Structure & Algorithms, System Programming, Computer Architecture, Artificial Intelligence, Computational Linguistics, Programming Languages & Compiler, Computer Networks, Analysis of Algorithms, Web Development*

WORK EXPERIENCE

Software Engineering Intern | Biaodian | Hangzhou, China Jan 2025 – Jul 2025

- Developed authentication system for delivery drivers using Java, Spring Boot, and Redis for session management on construction equipment rental platform
- Built a statistical reporting dashboard with MySQL queries and automated data generation for business intelligence
- Implemented Spring Cloud microservices with Feign clients for inter-service communication and Spring Filters for request validation
- Integrated Flowable workflow engine to automate contract and order signing processes
- Conducted migration from third-party driver management to in-house solution

Software Engineering Intern | Redvest | Los Angeles, CA Jun 2024 – Aug 2024

- Led a team of 6-7 interns as Tech Lead in developing Bankzzz, a platform connecting startup companies with potential users
- Architected and implemented a full-stack application using React, Tailwind CSS for frontend and [Node.js](#) accompanied with Google Firebase for backend services
- Oversaw project's GitHub repository and deployment pipeline using Firebase, including Firestore database integration
- Established development workflows and best practices for intern team, ensuring consistent code quality and timely delivery

Research Assistant/Software Engineer | Coordinated Science Laboratory | Champaign, IL Apr 2022 – May 2023

- Contributed in DARPA's INfluence Campaign Awareness and Sensemaking project, serving over 100 researchers
- Engineered a robust frontend using JavaScript, React, and Bootstrap for effective data visualization using libraries like d3, Plotly, and Chart.js
- Collaborated with 15 + member team to develop a Backend data post-processing pipeline with Python and Redis, streamlining ML model outputs for frontend integration

PROJECTS

Polyglot AI Document Pipeline | <https://prism-reader.space>

- Architected a distributed, polyglot microservices application using Go (Gin) for a high-concurrency API gateway and Python for asynchronous AI processing, fully containerized via Docker and deployed through Terraform
- Implemented an event-driven task queue using Redis and AWS SQS to decouple client-facing file ingestion from heavy LLM workloads, ensuring high availability during large PDF uploads to AWS S3
- Engineered a stateful AI worker workflow in Python to perform context-aware extraction and execute high-fidelity bilingual translations across large document contexts
- Built a highly interactive React + Vite.js client featuring a responsive split-screen UI, mapping translated text chunks back to original document coordinate data in real-time for a seamless bilingual reading experience

Cross-platform Stock Simulator App

- Deployed a real-time stock simulation platform on GCP App Engine integrating Finnhub API for live market data, with a Node.js/Express.js backend, MongoDB persistence, and RESTful APIs for portfolio tracking and trade execution
- Built an Angular/Bootstrap/Highcharts frontend featuring interactive stock charts, autocomplete search, and real-time portfolio monitoring dashboards; developed a native Swift iOS companion app with cross-platform feature parity
- Implemented end-to-end and unit tests to validate API endpoints and data pipeline reliability across web and mobile clients before each deployment

SKILL SET

Languages: Python, Go, Java, JavaScript, TypeScript, C, C++, Swift, SQL, HTML/CSS

Frameworks & Libraries: React, Angular, Next.js, Node.js, Express.js, Spring Boot, Gin, Flask, FastAPI, Tailwind CSS, MaterialUI

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Git, CI/CD, Terraform

Databases & Infrastructure: PostgreSQL, MongoDB, MySQL, Redis